

- 7 (a) (i) Sound cannot be polarised [B1] as it is longitudinal [B1].
- (ii) The particles are more tightly packed in solid steel [B1] than in air. Hence it is faster for energy to be transferred between particles.
- (iii) I is inversely proportional to square of r . $I \propto \frac{1}{r^2}$ [B1]

iv.1 $10 \lg \frac{I_x}{(1.0 \times 10^{-12})} = 120$

$I_x = 1.0 \text{ W m}^{-2}$ [B1]

$20 \lg \frac{p_x}{(20 \times 10^{-6})} = 120$

$p_x = 20 \text{ Pa}$ [B1]

iv.2 Estimated area of an eardrum = $1 \text{ cm}^2 = 10^{-4} \text{ m}^2$ [B1]

(accepted area between $2 \times 10^{-5} \text{ m}^2$ to $2 \times 10^{-4} \text{ m}^2$; based on circular area between diameter of 0.5 cm to 1.5 cm)

iv.3 $F = PA$
 $= 20(10^{-4})$ [C1]
 $= 2.0 \times 10^{-3} \text{ N}$ [A1]

- (b) (i) To avoid causing noise disturbances to people living on land. [B1]

OR

As no human lives on the ocean, no one will be affected by the noise disturbance. [B1]

- (ii) Any of the following (or other sensible suggestions): [B1]

Minimise the drag force at high speed as air is thinner at high altitude.

Minimal obstacle / traffic at high altitude so that it can travel at high speed without obstructions.

Minimise intensity of noise at sea level.

- (ii) Speed = $(1.4)(295) = 413 \text{ m s}^{-1}$ [B1]
- (c) (i) Resonance [B1]
- (ii) Wavelength
- $$= 2(0.91 - 0.54) \quad [\text{C1}]$$
- $$= 0.74 \text{ m} \quad [\text{A1}]$$
- (iii) Speed of sound = $f\lambda$
- $$= (480)(0.74) \quad [\text{C1}]$$
- $$= 355 \text{ m s}^{-1} \quad [\text{A1}]$$
- (iv)
$$v = 331 \sqrt{\frac{22 + 273}{273}} \quad [\text{M1}]$$
- $$= 344 \text{ m s}^{-1} \quad [\text{A1}]$$